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(Pages : 4)

Name.....

Reg. No.....

**FIFTH SEMESTER U.G. DEGREE [SPECIAL] EXAMINATION
NOVEMBER 2020**

(CUCBCSS—UG)

Mathematics

MAT 5B 05—VECTOR CALCULUS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 5B 05—VECTOR CALCULUS
(Multiple Choice Questions for SDE Candidates)

1. The angle between the vectors $a = [1, 2, 3]$ and $b = [0, -2, 1]$ is :
- (A) $\cos^{-1} \frac{1}{\sqrt{60}}$. (B) $\cos^{-1} \frac{-1}{\sqrt{70}}$.
(C) $\cos^{-1} \frac{1}{\sqrt{70}}$. (D) $\cos^{-1} \frac{1}{\sqrt{80}}$.
2. The straight line through the point $(1, 3)$ in the xy plane and perpendicular to the straight line $x - 2y + 2 = 0$ is :
- (A) $3x - y = 2$. (B) $x + y = 1$.
(C) $2x + y = 5$. (D) $2x - y = 5$.
3. The parametric equations for the line through $(-3, 2, -3)$ and $(1, -1, 4)$ are :
- (A) $x = 1 + 4t, y = -1 - 3t, z = 4 + 7t$. (B) $x = 2 + 4t, y = -2 - 3t, z = -4 + 7t$.
(C) $x = 3 + 4t, y = 8 - 3t, z = 5 + 7t$. (D) $x = 1 - 4t, y = -1 + 3t, z = -4 - 7t$.
4. The point of intersection of the line $x = \frac{8}{3} + 2t, y = -2t, z = 1 + t$ and the plane $3x + 2y + 6z = 6$ is :
- (A) $(1, 1, 2)$. (B) $(2, 0, 1)$.
(C) $(\frac{2}{3}, 2, 0)$. (D) $(0, 1, 3)$.
5. If $r(t) = \sin ti + e^{-t}y + 3k$, then $\frac{dr}{dt}$ is :
- (A) $\sin ti + 3k$. (B) $\cos ti + e^{-t}j + 3k$.
(C) $\cos ti - e^{-t}j$. (D) $\sin ti - e^{-t}j$.
6. The domain of the function $f(x, y, z) = xy \ln(z)$:
- (A) Entire Space (B) $\{(x, y, z) : xyz \neq 0\}$.
(C) Half space $z > 0$. (D) Half space $z < 0$.
7. Which of the following holds for the function $f(x, y) = \frac{4x^6y^2}{x^{12} + y^4}$?
- (A) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists. (B) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ doesn't exist.
(C) $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$. (D) None of these.

8. Let $f(x, y) = x - y$ and $g(z, y) = e^z$ be two continuous functions. Then the composition function $g(f(x, y)) = e^{x-y}$ is :
- (A) Discontinuous. (B) Continuous.
(C) Continuous at origin. (D) None of these.
9. The function $f(x, y) = xy$ has a :
- (A) Local maximum.
(B) Local minimum.
(C) Both local maximum and minimum.
(D) No local extreme values.
10. The minimum value that the function $f(x, y) = xy$ takes on the ellipse $\frac{x^2}{8} + \frac{y^2}{2} = 1$ is :
- (A) 2. (B) -2.
(C) 4. (D) -4.
11. The plane $x + y + z = 1$ cuts the cylinder $x^2 + y^2 = 1$ in an ellipse. The points on the ellipse that lies closest to the origin are :
- (A) $(1, 0, 0)$ and $(0, 0, 1)$. (B) $(0, 1, 0)$ and $(0, 0, 1)$.
(C) $(1, 0, 0)$ and $(0, 1, 0)$. (D) $(1, 0, 0)$ and $(0, 1, 1)$.
12. What is the value of $\iint xy dx dy$ over the first quadrant of the circle $x^2 + y^2 = a^2$?
- (A) $\frac{a^2}{4}$. (B) $\frac{a^2}{8}$.
(C) $\frac{a^4}{4}$. (D) $\frac{a^4}{8}$.
13. A coil spring lies along the helix $r(t) = (\cos 4t) \mathbf{i} + (\sin 4t) \mathbf{j} + k, 0 \leq t \leq 2\pi$. The spring's density is a constant, $\delta = 1$. Then the radius of gyration of the spring about the z -axis is :
- (A) 1. (B) 2.
(C) 3. (D) 4.
14. The gradient field of $f(x, y, z) = xyz$ is :
- (A) $yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$. (B) $xy\mathbf{i} + xz\mathbf{j} + yz\mathbf{k}$.
(C) $xz\mathbf{i} + yz\mathbf{j} + xy\mathbf{k}$. (D) None of these.

15. If $\nabla\phi = (y + y^2 + z^2)i + (x + z + 2xy)j + (y + 2xz)k$ and $\phi(1, 1, 1) = 3$, then what is ϕ ?

(A) $xz + xy + yz^2 - 1.$

(B) $xz + yz + xz^2.$

(C) $xy^2 + xz^2 - 1.$

(D) $xy + xy^2 + xz^2 + yz - 1.$

16. Which among the following is the work done in moving a particle once round a circle C in the xy-plane. Given the circle has centre at the origin and radius 3 and the force field is given by

$$F = (2x - y + z)i + (x + y - z^2)j + (3x - 2y + 4z)k.$$

(A) $8\pi.$

(B) $80\pi.$

(C) $88\pi.$

(D) $18\pi.$

17. If $F = (3x^2 + 6y)j - 14yzj + 20xz^2k$, then the value of $\int_C F \cdot dr$ where C is a curve from

$(0, 0, 0)$ to $(1, 1, 1)$ with parametric from $x = t, y = t^2, z = t^3$ is :

(A) 13.

(B) 7.

(C) 5.

(D) 11.

18. If $\hat{r}$ is the unit vector in the direction of r and $r = |r|$, then $\text{div}(\hat{r})$ is :

(A) $\frac{r}{2}.$

(B) $r.$

(C) $zr.$

(D) $\frac{z}{r}.$

19. Vector product is :

(A) Commutative.

(B) Anticommutative.

(C) Associative.

(D) Not distributive wet vector addition.

20. If F,G are differentiable vector functions and ϕ is a differentiable scalar function. Then :

(A) $\text{curl}(F \times G) = (\text{grad } \phi) \times F + \phi \text{curl}(F).$

(B) $\text{div}(F \times G) = -F \text{curl } G + G \text{curl } F.$

(C) $\text{div}(F \times G) = (G, \nabla)F - (F, \nabla)G + F \text{div } G - G \text{div } F.$

(D) $\text{curl}(F \times G) = F \text{curl } G - G \text{curl } F.$